

Saran Citizen WhatsApp Bot: Detailed Implementation Plan

Phase 1: Sonpur Mela

Version 1.3 · 25 Sep 2026 · M0–M3 built · Phase 1 = Sonpur Mela only; local-first hosting with VPS fallback · Branch `claude/whatsapp-n8n-llm-chatbot-o9gtyo`

Summary.

- **What:** a WhatsApp bot built on n8n where a citizen types "Hi" and gets **Sonpur Mela** information: programme, control room, police, health and vet camps, parking, ghats.
- **Data:** it answers from Google Sheets that staff maintain, never from LLM guesses.
- **Near me:** a citizen can share their location to get the nearest police station, health centre or vet camp, with walking directions.
- **Growth:** the **Officer Directory** and **Scheme Eligibility** are added later, with DM approval. They plug in by adding a service row and a workflow, without rebuilding the bot.
- **Hosting:** the office machine first (behind a Cloudflare Tunnel), with a pre-arranged Hostinger-type VPS as fallback. Measured go/no-go criteria decide the move.
- **Schedule:** go-live is targeted around **10 Nov 2026**, two weeks before the Mela opens, after three test-gated milestones.

0. Context

What exists today. The repo has a working prototype of a **Sonpur-Mela-only** bot:

- n8n + local Ollama + Postgres/pgvector
- CSV data loaded by `scripts/ingest.py`
- one monolithic workflow, `n8n/workflows/sonpur-mela-bot.json`, with keyword routing

The goal. Turn the prototype into a **citizen-services WhatsApp platform**. A citizen sends any message and is served by the enabled services.

Phase 1 scope: one service, Sonpur Mela.

The District/Block Officer Directory (design ready, sample analysed) **and Scheme Eligibility** come in later phases, **after DM sir's approval**. They plug into the same core without rebuilding it (see §14).

More services must plug in later without rebuilding the chatbot. Every interaction is logged for analytics and feedback.

Confirmed decisions

- **Build order:** Core platform → Sonpur Mela → Go-live readiness → Analytics, with a test gate after each milestone. After that: Directory, then Schemes.
- **LLM:** hosted Claude Haiku 4.5 (`claude-haiku-4-5`). Ollama and pgvector are retired.
- **Languages:** Hindi + English.
- **Hosting:** the **office machine first** (Cloudflare Tunnel, §2), with a **Hostinger-type private VPS** as the pre-arranged fallback (§2.4).
 - A private VPS is acceptable because the Mela content is public-domain information.

- Citizen phone numbers and messages remain personal data, so they stay hashed/encrypted with limited retention on any host (§7).
- The user chose local-first over the recommendation of running production on the VPS; §2.4's criteria make that safe.
- **WhatsApp:** the production number is live and verified.
- **Duty-staff mobile numbers:** approved for publishing.
- **Data format:** the samples are indicative. The bot reads the recommended template format in §5b.
- **Data editing:** Google Sheets is where staff edit data. It is synced into Postgres every 10 min; the bot never reads Sheets live.

1. Senior-developer review findings

1a. Defects in the existing code (verified by reading it)

#	Finding	Evidence	Fix
E1	Only the first message of a webhook is processed	<code>Extract Message</code> reads <code>entry[0].changes[0].messages[0]</code> . Meta can batch several entries/messages into one POST	Router splits <code>all entry[] × changes[] × messages[]</code> into separate items
E2	No webhook signature check: anyone who knows the URL can POST fake messages, spend LLM credits and spam replies	No <code>X-Hub-Signature-256</code> check anywhere	Verify HMAC-SHA256 of the raw body with <code>META_APP_SECRET</code> ; reject on mismatch
E3	Graph API pinned to v19.0	Send WhatsApp Reply URL	[Likely] v19 is at or near Meta's ~2-year deprecation. Make it <code>GRAPH_API_VERSION</code> in env and set it to the current version at build time
E4	<code>N8N_BASIC_AUTH_*</code> in compose	<code>docker-compose.yml</code>	[Likely] Removed in n8n 1.x, which uses built-in user management. The editor is effectively protected only by the owner account and is exposed on 0.0.0.0:5678. Fix: bind to 127.0.0.1, expose only <code>/webhook/*</code> through the Cloudflare Tunnel, and reach the editor over Tailscale
E5	<code>n8nio/n8n:latest</code> , <code>ollama:latest</code>	compose	Pin exact versions. A silent n8n upgrade mid-Mela can break node schemas
E6	No <code>N8N_ENCRYPTION_KEY</code> set	compose	If the n8n volume is lost, all stored credentials become unreadable. Set it explicitly and back it up
E7	Workflow JSON has no fixed id	<code>id: None</code> in the export	Required for registry dispatch (see R1)
E8	Non-text messages (voice notes, images) get a silent ack	Ack <code>Non-Text</code>	Voice notes will be common. Reply in Hindi: "Please type or use the menu" and log the type (voice-to-text is a candidate future service)

1b. Design risks fixed in this plan

#	Risk	Fix
R1	n8n assigns new workflow IDs on import, which breaks dispatch-by-ID; hand-edited workflow JSON can't be unit-tested	Workflows are generated from source (<code>n8n/src → scripts/build_workflows.mjs</code>) with fixed ids; Code-node logic is unit-tested JS. Deploy with <code>scripts/n8n_import.sh</code> (import + publish). The UI is for emergencies only (<code>scripts/n8n_export.sh</code>)

#	Risk	Fix
R2	Meta status callbacks (sent/delivered/read) run about 3× the inbound volume	Dropped at the first node. <code>EXECUTIONS_DATA_SAVE_ON_SUCCESS=none</code> , prune >72h. <code>core.message_log</code> is the audit trail
R3	Stale buttons tapped after the session expired	Self-describing IDs <code>svc:action:arg</code> (e.g. <code>mela:fac:medical</code>). Routing by ID prefix works without session state
R4	WhatsApp field limits: button title ≤20, list row title ≤24, row description ≤72, list button ≤20, ≤3 buttons, ≤10 rows, interactive body ≤1024, text ≤4096	Meta rejects the whole message if any one field is over. The send workflow validates and truncates, and <code>CI check_templates.py</code> fails on overflow in hi or en
R5	Admin alerts and the daily report are business-initiated, so they are blocked outside the 24h window	An approved utility template <code>admin_alert</code> (1 variable), plus email as backup
R6	Non-atomic sync can expose half-loaded tables	One SQL function per table does delete+insert in a single transaction
R7	The in-charge roster would need facilities × 32 days × 3 shifts ≈ thousands of rows	Confirmed by the sample: the roster is a standing 3-shift pattern (the same person every day). <code>duty</code> has no date; a blank shift means all-day
R8	Concurrent messages from one user race on session state	<code>sessions.version</code> optimistic lock, with one retry
R9	Sub-workflows have no HTTP entry, so they can't be tested	<code>core/99-test-harness</code> webhook, active only when <code>ENV=test</code>
R10	Load tests would hit the real Claude API and Meta	<code>ANTHROPIC_BASE_URL</code> and <code>WA_API_BASE</code> point at local mocks in test mode
R11	Prompt caching won't help	Haiku 4.5's minimum cacheable prefix is 4096 tokens and <code>history.md</code> is smaller. No caching; cost is negligible
R12	A language-picker first step causes drop-off	Auto-detect (Devanagari → hi, otherwise en). Every menu has a "🌐 English/हिंदी" toggle
R13	Sending citizen text to a foreign API needs departmental approval	Get written sign-off in week 1. <code>LLM_ENABLED=false</code> gives a fully working menu-only bot. The phone number is never sent to the LLM
R14	Spam/abuse, or a loop with another bot	Per-user rate limit: >20 inbound per 5 min → one "please slow down" reply, then ignore for 10 min
R15	No error path: a failed execution means the citizen gets silence	An n8n Error Trigger workflow sends the citizen a "Sorry, please try again / menu" reply and an <code>admin_alert</code> to staff
R16	No monitoring	An external uptime check on <code>/healthz</code> (tunnel route to an n8n health webhook, checked by UptimeRobot every 5 min with SMS/email alerts), a daily sync-freshness check, and disk-usage alerts
R17	A "zero core changes" gate is too absolute	Gate = no core <i>feature</i> changes to add a service. Core bug fixes go in separate commits with tests

2. Architecture (office machine first, VPS fallback)

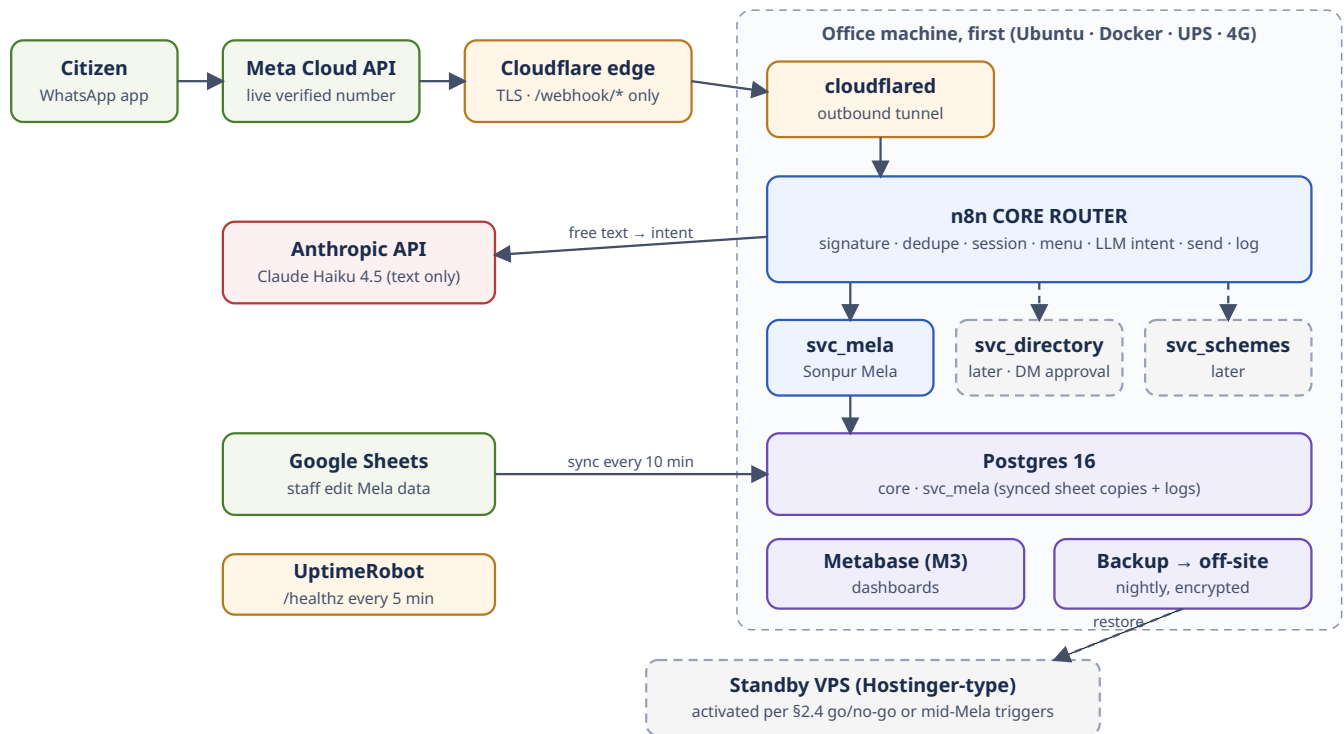


Figure 1. Phase 1 architecture: core router + Sonpur Mela service on the office machine, with later services and the standby VPS shown dashed.

2.1 Why Cloudflare Tunnel instead of port-forwarding/Caddy

[Certain] Meta requires a public HTTPS URL with a valid certificate. An office machine usually has no static IP, sits behind NAT, and has an IT firewall.

- **cloudflared connects outbound only**, so there's no static IP, no port forwarding, and no inbound firewall rule. It gets a real TLS certificate on `bot.<domain>`, and Cloudflare Access rules block every path except `/webhook/*` and `/healthz`.
- **Cost**: free tier. It needs a domain (or subdomain) whose DNS is on Cloudflare.
- **Fallback**: if IT forbids Cloudflare, use port-forward 443 + Caddy + a static IP from the ISP (≈₹500–1,500/month extra). The compose file keeps this as a `caddy` profile.

2.2 Office-machine requirements (on-prem is now the #1 availability risk)

Item	Minimum	Why
CPU / RAM / disk	4 cores, 16 GB RAM , 256 GB SSD	n8n + Postgres + Metabase + OS headroom (no local LLM)
OS	Ubuntu 24.04 LTS (dedicated, not someone's daily PC)	Docker restart policies; no Windows updates rebooting it mid-Mela
Power	Online UPS ≥30 min + building inverter/genset	Mela month in Nov–Dec; power cuts are likely [Likely]
Internet	Primary broadband + 4G/5G dongle or router failover	Meta retries for a limited time only; long outages mean lost replies
Boot	BIOS "power on after AC loss"; Docker + cloudflared as systemd services	A reboot must recover by itself with no one on site
Remote access	Tailscale on the machine and admin laptops	Night-time fixes without visiting the office

Item	Minimum	Why
Physical	Locked room, labelled "do not switch off"	The most common real-world on-prem outage cause [Likely]

2.3 Disaster recovery (cold standby)

- **Nightly backup:** `pg_dump` + the env/keys file, encrypted, copied off-site.
- `docs/runbook.md` → **"Move to cloud in 1 hour":**
 1. Rent any VPS.
 2. `git clone`.
 3. Restore the dump.
 4. Install `cloudflared` with the same tunnel token.

The DNS and the Meta webhook URL stay the same, so citizens notice nothing. **This is rehearsed once before go-live** (M2 gate). It is automated by `scripts/provision_vps.sh`; the full criteria and procedure are in §2.4.

Component versions (pinned in compose): n8n **2.40.5** (needs Node ≥ 24) · postgres 16.13 · caddy 2.10 · cloudflared · metabase (M3).

Environments. Two compose projects run on the same machine:

- **staging:** Meta test number, `staging.<domain>` tunnel route, `ENV=staging`.
- **prod:** the live number, `bot.<domain>`, `ENV=prod`.

Each has its own DB and env file. Promotion from staging to prod imports the same workflow JSON files.

2.4 Fallback to a VPS: criteria, standby, switch

Burn-in. The office machine runs the staging stack **24×7 from the start of M1** (about 4 weeks before go-live). UptimeRobot records uptime from day one, so the decision rests on measured data rather than impressions.

Go/no-go at the M2 gate ($\approx$ 3 Nov). Go live on the VPS instead if **any** of these fails:

#	Criterion (over the burn-in)	Fail if
H1	<code>/healthz</code> uptime (UptimeRobot)	< 99.5% ($\approx$ more than 3.5 h down in 4 weeks)
H2	Longest single outage	> 30 min
H3	Pull the power cable, then restore power	Needs a human to recover
H4	Pull the primary ISP cable	4G failover doesn't restore the webhook within 5 min
H5	k6 load test on the machine	p95 > 3 s at 30 msg/s
H6	Office constraints	IT blocks outbound HTTPS to Cloudflare/Meta/Anthropic/Google, or the machine is shared or switched off at night

Standby VPS

- **What:** a Hostinger (or similar) KVM VPS with 2+ vCPU, 8 GB RAM, and an **India data centre if offered** [Likely: Hostinger lists one; verify at purchase]. Cost $\approx$ ₹700–1,500/month [Guessing: check current pricing]. It is bought only for the M2 drill and on activation.
- `scripts/provision_vps.sh`: one idempotent script that:
 1. installs Docker, ufw, fail2ban, Tailscale, and unattended-upgrades

2. installs either the same `cloudflared` tunnel token or Caddy + Let's Encrypt
3. restores the latest off-site backup
4. imports the workflows

It is tested once in M2 on a throwaway VPS.

- **The webhook URL (`bot.<domain>`) doesn't change**, so **Meta needs no re-configuration**. Only the tunnel/DNS target moves.

Mid-Mela triggers (after go-live on the office machine). Move to the VPS **the same day** if:

- a single outage lasts > **1 hour**, or
- there are **3 or more outages in 24 h**, or
- the office or machine can't be reached for maintenance.

The runbook procedure takes about 1 hour. The named technical owner decides; the move is pre-approved by this plan.

Recorded decision. The recommendation was "VPS for production, office machine for staging", which removes the #1 outage risk for about ₹1k/month. The user chose local-first. This can be revisited at the go/no-go without any code change.

3. Core platform design

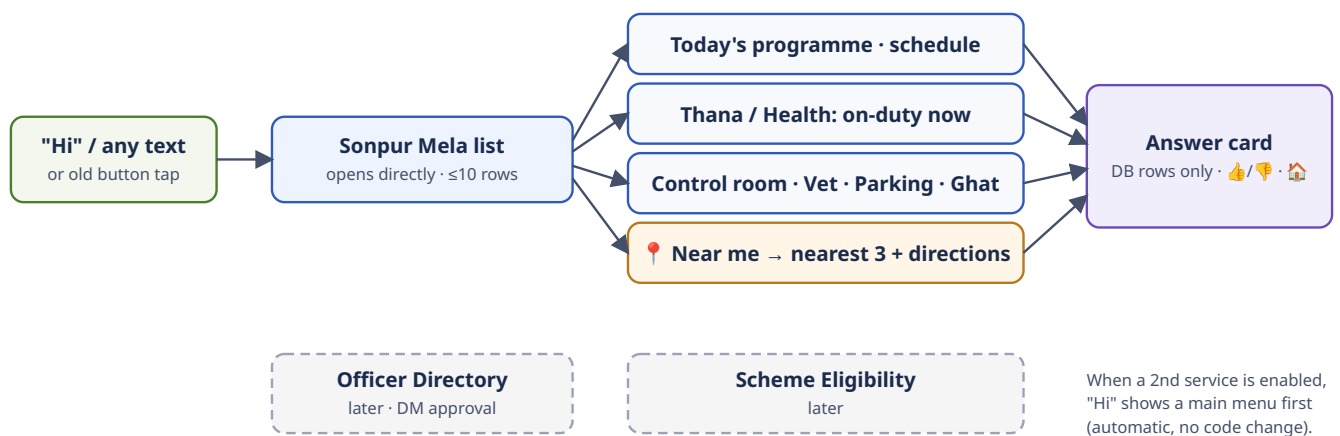


Figure 2. Citizen journey in Phase 1. Every answer comes from verified sheet data, never from LLM text.

3.1 Router pipeline (`core/00-router`, node by node)

1. **Webhook POST** `/webhook/wa`. Raw body enabled; responds **200 immediately** (Meta retries on a slow ack).
2. **Verify signature** (Code; `NODE_FUNCTION_ALLOW_BUILTIN=crypto`). On mismatch: log a security event and stop.
3. **Explode + filter** (Code). Walks all entries, changes, and messages. Drops `statuses`. Output: one item per message.
4. **Begin turn** (Postgres: `core.begin_turn(wa_id, msg_id, ts, type)`), a single round trip that:
 - HMACs the phone number with `PHONE_HASH_SECRET`
 - dedupes (`INSERT ... ON CONFLICT DO NOTHING`)
 - upserts the citizen (the number is stored `pgp_sym_encrypt`-ed)
 - checks the rate limit
 - loads the session and its version
 - returns `{is_dup, rate_limited, wa_hash, lang, state, context, version, first_contact}`
5. **Stop** if `is_dup`. If `rate_limited`, send one notice and stop.

6. **Normalize input** (Code). Produces `kind` (text/button/list/location/unsupported), `id`, `text`, `lat/lon`, and the detected language.
 7. **Resolve route** (Code), in priority order:
 - a global command (`hi|hello|namaste|menu|0|start|मेनू|help`) → core menu
 - a `lang:* id` → switch language
 - an `fb:* id` → feedback
 - an `id` prefix matching a registry `id_prefix` → that service
 - a session state prefix → that service
 - free text → LLM classify
 - otherwise → menu
 8. **LLM classify** (`core/02-llm`, only for free text with no active flow expecting text). A confident result becomes a service route with a synthetic `input.intent`. Low confidence or an error sends the menu and logs to `unanswered`.
 9. **Dispatch** (Switch → Execute Workflow; the workflow ID comes from the registry row, "wait for completion" on).
 10. **Validate output** (Code). Checks the contract schema. Invalid output → error path.
 11. **Decorate** (Code). On the first contact, the privacy notice is prepended. If `done=true`, the feedback buttons `[👍][👎][🏠 Menu]` are appended.
 12. **Send** (`core/01-send`, per message):
 - render the hi/en template
 - validate limits (R4)
 - **phone guard**: every phone-like token must exist in `core.allowed_numbers` (a view: union of service phones + 112/108/1098)
 - POST to Graph API
 - log the outbound message
 - retry once on 5xx/429
 13. **End turn** (Postgres: `core.end_turn(wa_hash, version, next_state, context, log)`). On a version conflict, re-run once from step 6.
- **Error Trigger** (`core/08-error`). Sends the apology + menu to the citizen and an `admin_alert` with the execution ID.

3.2 Service contract

```

INPUT  { wa_hash, lang: "hi"|"en", state, context, now_ist, is_admin,
         input: { kind, id, text, lat, lon, intent?: {subtype, slots} } }
OUTPUT { messages: [ {type:"text", body} |
                     {type:"buttons", body, buttons:[{id,title}] (≤3)} |
                     {type:"list", body, button, sections:[{title, rows:[{id,title,description}]}] (≤10
rows)}} |
         {type:"location", lat, lon, name, address} |           // map pin, opens in Maps
         {type:"location_request", body} ],                     // "Send location" button
         next_state, context, done, log: { subtype, resolved } }
```

Rules:

- Services never call Meta or write to `core.*`.
- Every state and every ID starts with the service's `id_prefix`.
- A service must handle any stale ID of its own gracefully.

3.3 Registry (`core.services`)

`service_key` PK, `id_prefix` UNIQUE, `title_hi`, `title_en` (≤20), `description_hi/en` (≤72), `menu_order`, `enabled`, `workflow_id`, `intent_hint_en`, `intent_hint_hi`, `subtypes` jsonb {key: hint}, `keywords` text[].

- **Main menu.**

- **Exactly 1 enabled service** (Phase 1, Mela only): "Hi" goes **straight to that service's menu**, with no one-button main menu.
- 2–3 services: reply buttons.
- More than 3: a list, where the row description comes from `description_*`.
- Enabling the Directory later brings the main menu back automatically.
- **LLM prompt.** Built from `services` + `subtypes` at call time, so new services are classified automatically.

3.4 LLM specification (`core/02-llm`, HTTP Request → `POST /v1/messages`)

Classifier

- **Request:** `claude-haiku-4-5`, `max_tokens 300`, `temperature 0`, `timeout 8s`. The system prompt is generated from the registry.
- **Output:** `output_config.format` JSON schema: `{service_key: enum[...enabled keys, "none"], subtype: string, slots: {block, department, date, facility_type}, lang: enum[hi,en], confidence: number}`
- **Threshold:** accept if `confidence ≥ 0.6` and `service_key ≠ none`.

Mela Q&A (used by `svc_mela` via the same sub-workflow with `mode=qa`)

- **System prompt:** `history.md` + do's & don'ts, loaded from the DB (`svc_mela.qa_context`).
- **Output:** `{answerable: bool, answer: string ≤600 chars}`, in the user's language, with no phone numbers.
- **Fallback:** if `answerable=false`, the bot says it doesn't know and shows the Mela menu.

General

- **Logging:** every call records `llm_tokens` and latency.
- **Kill switch:** with `LLM_ENABLED=false`, both calls are skipped.

3.5 Conversation rules

- **Session timeout:** 30 min idle. The next message starts at the menu, but self-describing IDs still work.
- **Menu shortcuts:** "0" or "menu" works anywhere. Each result ends with the feedback + menu buttons.
- **Unsupported message types** (voice, image, video, sticker, document): the bot replies with a template and shows the menu.
- **Location messages:** routed to the active service if it declares `accepts_location`. Otherwise they go to the service with a pending location request, and if there is none, to `svc_mela`'s "nearest" handler during the Mela window (§6.1a) or to the menu outside it.
- **Admins:** `core.admins` (hashed numbers, seeded from `ADMIN_WA_NUMBERS`) sets `is_admin` in the service input. Admin-only IDs (`*:adm:*`) are rejected for everyone else.

4. Data model (`sql/*.sql`, applied in order; idempotent)

core schema (`00_core.sql`)

- `services` as in 3.3.
- `templates` (key PK, hi, en, max_len).
- `citizens` (wa_hash PK, wa_number_enc bytea, lang, first_seen, last_seen, notice_shown_at, msg_count).
- `sessions` (wa_hash PK, state, context jsonb, lang, version int, updated_at).
- `message_log` (id bigserial, wa_msg_id text UNIQUE NULL, wa_hash, direction in/out, service_key, state, subtype, kind, text, via button|list|text|llm|system, resolved, llm_tokens, latency_ms, error, ts).
 - Indexes: (ts) and (service_key, ts).
- `feedback` (id, wa_hash, message_log_id, rating smallint, ts).
- `unanswered` (id, wa_hash, text, service_key, reason, ts).
- `sync_runs` (id, service_key, source, status ok|rejected|error, rows, errors jsonb, ts).

- **rate_limit**: handled inside `begin_turn` with a count over `message_log` in the last 5 min (indexed), so no separate table.
- **Functions**: `begin_turn`, `end_turn`, `purge_older_than(days)`.
- **View allowed_numbers**, plus analytics views `v_daily`, `v_top_subtypes`, `v_feedback`. These expose no raw numbers.

svc_mela schema (10_svc_mela.sql), shaped by the real 2025 sample

- **shifts** (`shift_no 1..3`, `label`, `start_time`, `end_time`).
 - Seeded 06:00–14:00, 14:00–22:00, and 22:00–06:00; this pattern is identical in the Control Room and Thana tabs.
 - `current_shift(now_ist)` handles Shift-3 crossing midnight.
- **places** (`id`, `category`, `name_en`, `name_hi NULL`, `lat NULL`, `lon NULL`, `sort`).
 - `lat/lon` come from the sheet.
- **place_coords** (`place_id PK`, `lat`, `lon`, `captured_by_hash`, `captured_at`): coordinates captured on-site by admins (§6.1b).
 - **The sync never touches this table.**
 - Effective location = `COALESCE(sheet lat/lon, captured lat/lon)`, so a value typed into the sheet overrides a capture.
 - The view `v_places_geo` exposes it.
 - `category` is one of: `health_centre`, `thana`, `vet_camp`, `parking`, `ghat`, `accommodation`, `other`.
- **duty** (`place_id`, `shift_no NULL`, `person_name`, `phones text[]`, `role NULL`).
 - `shift_no NULL` means all-day. Vet camps have no shifts. Merged cells in the source mean the same person on every shift.
 - `phones` is an array, because source cells hold "8709898293 & 7970663565".
- **control_room_desks** (`desk_en`, `desk_hi`, `sort`) and **control_room_duty** (`desk`, `shift_no NULL`, `person_name`, `designation`, `phones text[]`).
 - The source is a desk × shift grid: Magistrate, Police, Sanitation & Water, Electricity, Health.
- **control_room_public** (`label`, `phone`): the public helpline number(s). **Missing from the sample** (see §5a).
- **events** (`date`, `department`, `programme_en`, `programme_hi NULL`, `time NULL`, `venue NULL`).
 - The sample has only date, department, and artist/programme.
- **guidelines** (`kind do|dont`, `text_hi`, `text_en`, `sort`). **Not in the sample.** The menu row is hidden until rows exist.
- **qa_context** (`id=1`, `content`): `history.md` + guidelines.
- **Functions**: a `replace_<table>(jsonb)` function per table, `current_shift()`, and `on_duty(category, now_ist)`.
- **nearest(category, lat, lon, n=3)**: haversine distance over `v_places_geo`, limited to `active AND verified` `AND` coordinates present.
 - Returns `place`, `distance_m`, `on-duty person`, `phone`.
 - Plain SQL is enough for about 50 sites, so no PostGIS is needed.

5. Data: what the samples show, and the sheet format the bot will read

5a. Findings from the Mela sample (Sonpur_Mela_Data.xlsx)

#	Finding	Evidence	Consequence / action
D1	The files are print layouts, not data tables. They use merged title rows, two-row headers, and shifts as column groups; the directory sample also stacks several sections in one tab	Every tab	A sync can't reliably parse these. Decision: we convert them once into a "bot format" Google Sheet: one tab per table, one header row, no merged cells. Staff keep editing that sheet (§5b). The one-time conversion script <code>scripts/convert_samples.py</code> is part of M1/M2

#	Finding	Evidence	Consequence / action
D2	Legacy Hindi font encoding	Health centre "Nakhas ¼u[kk'k½" is Kruti Dev text, not Unicode, so it would reach citizens as garbage	The sync rejects cells containing Kruti Dev markers (¼ ½ [k patterns). Hindi must be typed in Unicode. The converter strips the part and flags it
D3	Invalid phone	Thana "Sawaich Ghat" Shift-2: 709095094 (9 digits)	Validator: mobiles must be 10 digits starting 6–9; landlines need an STD code with a leading 0. The row is rejected with its row number in the alert
D4	A landline stored without its leading 0 looks like a mobile	Seen in the directory sample: Forest Division Officer 6152232660 (Chapra STD 06152). The same risk applies to any Mela landline	Excel stores numbers as integers and drops the leading 0. The phone column must be plain text in the sheet. 6152... / 6158... patterns are flagged for manual check
D5	Multiple people/phones in one cell	Control Room Sanitation & Electricity desks hold 2 officers and "8709898293 & 7970663565"	phones is stored as an array. The bot format has separate phone_1 and phone_2 columns
D6	Vertical merged cells mean "same for all shifts"	Control Room G4:G6, H4:H6, I4:I6	Converter: the value applies to shifts 1–3. Bot format: a blank shift means all-day
D7	Suspected copy-paste errors	Mela: "Bahadur Prasad Yadav" is on duty at two thanas with different numbers. (The directory sample has more: shared phone/email between different officers.)	A sync warning (not a reject) lists duplicate phones/emails across different people; the data owner confirms
D8	The Control Room tab lists internal duty officers but no public helpline number	The tab has no "call this number" line	Citizens need one number. Ask the district for the public control room number ; it goes in control_room_public. The desk roster is shown underneath
D9	Accommodation and do's & don'ts are not in the sample; Ghats are (6)	Sheet list	Mela menu rows are data-driven : a category with 0 rows is hidden automatically. Ghats become a menu row
D10	No lat/lon for any site	Blank columns	The "📍 Near me " feature (§6.1a) depends entirely on coordinates . The temporary sites only exist once they are set up, so coordinates are captured on-site by staff through the bot's admin tool (§6.1b) in the set-up week. Until then the feature switches itself off per category
D11	Schedule is 2025 (22 Nov–7 Dec 2025, with gaps on 4 and 6 Dec). No time or venue. Artist names are comma-joined	Mela Schedule tab	Good staging seed. The sync warns if no event date falls in the next 30 days. "Today" with no row → "No programme listed today" + the next listed date
D13	All data is English only	Both samples	The bot's own text is in hi/en. Names and places stay in English unless staff fill the optional *_hi columns. We won't machine-transliterate personal names , because wrong Hindi names on an official channel are worse than English ones
D14	Long names exceed WhatsApp list limits	"Veterinary Hospital Mobile Ambulance Van, Dist. Animal Husbandry Office" (74 chars, the list-row title limit is 24)	Places are listed in text cards, not list rows . Lists are used only for categories, blocks, and pages

#	Finding	Evidence	Consequence / action
D15	Mela duty staff numbers are personal mobiles (doctors, SIs)	Health/Thana tabs	Decided: publishing is approved. Keep a copy of the approval order in docs/approvals/. A <code>settings.publish_duty_phones</code> flag (default TRUE) allows an instant switch to names + helpline if an officer objects

5b. Recommended data format (the samples may change, so this is the target format)

Design rule: keep the sheet **as close as possible to how staff already write the duty orders**, one row per place with shift columns. Remove only the things that break machines: merged cells, two-row headers, several tables stacked in one tab, numbers stored as numbers, and legacy fonts. The sync (not the staff) turns this into normalised DB tables. Foreign-key tabs (a separate `duty` tab pointing at `place_id`) were rejected because staff mis-type IDs.

Ten rules for every tab (printed on a frozen `README` tab in each spreadsheet):

1. Row 1 holds the headers exactly as given. Never rename, merge, or reorder them; adding columns at the end is fine.
2. One table per tab. There are no title rows and no blank rows in the middle.
3. Phone columns are **Plain text** format. Use a 10-digit mobile (9431012345), or a landline with its STD code and a leading 0 (06152-240001). There is one number per cell; a second number goes in `phone_2`.
4. Dates are YYYY-MM-DD and times are 24-hour HH:MM.
5. Hindi is typed in **Unicode** (Google Input Tools or the Windows Hindi keyboard), **never Kruti Dev**.
6. Dropdown columns accept only the listed values.
7. `active` is TRUE/FALSE. Set it to FALSE to hide a row instead of deleting it; this keeps history.
8. `verified` is ticked only after the number has been called or checked against the order. Unticked rows never reach citizens in production.
9. Each tab has an `updated_by` column for the staff member's name.
10. The `id` column is a short unique code (e.g. TH01, HC03) and must never be reused.

Spreadsheet 1: "Sonpur Mela – Bot Data"

Tab	Columns (header row, exactly)	One row =
README	the rules above + a contact for the data owner	n/a
settings	key, value → <code>mela_start</code> , <code>mela_end</code> , <code>shift1_start</code> 06:00, <code>shift2_start</code> 14:00, <code>shift3_start</code> 22:00, <code>public_helpline_1</code> , <code>public_helpline_2</code> , <code>mela_center_lat</code> , <code>mela_center_lon</code> , <code>mela_radius_km</code> (default 5)	one setting
places	<code>id</code> , <code>category</code> ▼, <code>name_en</code> , <code>name_hi</code> , <code>location_en</code> , <code>location_hi</code> , <code>lat</code> , <code>lon</code> , <code>hours</code> , <code>s1_name</code>, <code>s1_phone</code>, <code>s2_name</code>, <code>s2_phone</code>, <code>s3_name</code>, <code>s3_phone</code>, <code>allday_name</code>, <code>allday_phone</code>, <code>phone_2</code> , <code>notes</code> , <code>active</code> , <code>verified</code> , <code>updated_by</code>	one thana / health centre / vet camp / parking / ghat / accommodation / help desk. Shift columns are blank where not applicable (parking, ghat)
control_room	<code>desk</code> ▼ (Magistrate, Police, Sanitation & Water, Electricity, Health, other), <code>s1_name</code> , <code>s1_designation</code> , <code>s1_phone</code> , <code>s2_...</code> , <code>s3_...</code> , <code>allday_name</code> , <code>allday_designation</code> , <code>allday_phone</code> , <code>phone_2</code> , <code>active</code> , <code>verified</code> , <code>updated_by</code>	one desk (the sample's grid maps 1:1)
events	<code>id</code> , <code>date</code> , <code>time</code> , <code>programme_en</code> , <code>programme_hi</code> , <code>artists</code> , <code>department</code> , <code>venue</code> , <code>is_highlight</code> , <code>active</code> , <code>verified</code> , <code>updated_by</code>	one programme item (several per day allowed)
guidelines	<code>id</code> , <code>kind</code> ▼ (do, dont, emergency), <code>text_en</code> , <code>text_hi</code> , <code>sort</code> , <code>active</code> , <code>verified</code> , <code>updated_by</code>	one instruction
faq (optional)	<code>id</code> , <code>question_en</code> , <code>question_hi</code> , <code>answer_en</code> , <code>answer_hi</code> , <code>active</code> , <code>verified</code>	an approved Q&A; fed to the LLM context and matched exactly first

category▼ values: thana, health_centre, vet_camp, parking, ghat, accommodation, toilet_water, lost_found, help_desk. **Adding a new category needs no code:** it shows up in the menu automatically once the category has at least one active, verified row. Its menu label comes from the category_labels rows in settings.

Example rows (from your sample, cleaned):

```
places: TH01 | thana | Nakhas Thana | नखास थाना | ... | Vishal Anand | 9308642005 | Rakesh Kumar |
6207037337 | Srijan Mishra | 8252877610 | | | TRUE | TRUE | Ramesh
places: VC08 | vet_camp | Veterinary Hospital, Sonpur | ... | (s1-s3 blank) | Dr. Vijay Prasad Mandal |
9155688071 | ... | TRUE | TRUE | Ramesh
control: Sanitation & Water | ... | allday_name: Shri Nikhil Kumar, AE PHED Chhapra | allday_phone:
8709898293 | phone_2: 7970663565
```

Deliverables for this format (built in M1/M2):

- data/templates/Sonpur_Mela_Bot_Data.xlsx (the Directory template follows in its phase), with dropdowns, plain-text phone columns, frozen headers, the README tab, and your sample already converted in. You upload them to Google Drive as Sheets.
- scripts/convert_samples.py, which converts the print-layout files you already have into the template, plus a flag report.
- docs/data-entry-guide.md (hi + en, 2 pages) for the data owners.

Validation (run by the sync every 10 min):

- **Errors:** missing required fields, invalid dropdown values, phone rules (D3/D4), Kruti Dev text (D2), duplicate id, or a shift name without a phone. **Result:** the tab is not updated, the last good data is kept, and one admin_alert per hour lists the tab and row numbers.
- **Warnings:** the same phone on different people (D7), no events in the next 30 days (D11), or last_verified older than 60 days. **Result:** they go into the daily digest.
- **Production:** only rows with active = TRUE and verified = TRUE are published.

6. Service designs

6.1 svc_mela (id_prefix mela). The menu is built from the data (empty categories are hidden)

Row id	Title (en / hi)	Answer
mela:today	Today's programme / आज का कार्यक्रम	Today's events. If none: "No programme listed today; next: <date> - <programme>". Ends with a [17 Full schedule] button (mela:schedule), which frees a list row for 📍 Near me and keeps the menu within the 10-row limit
mela:near	📍 Near me / मेरे पास	Asks which service (thana / health / vet / parking / ghat / control), then sends a "Send location" request (§6.1a). Shown only once at least one category has coordinates
mela:control	Control room / कंट्रोल रूम	Public helpline number(s) first, then on-duty desk officers for the current shift (Magistrate, Police, Sanitation & Water, Electricity, Health)
mela:cat:thana	Police stations / पुलिस थाना	All 14 temporary thanas with the current-shift in-charge + phone in one card (≈1.2k chars), with a shift label and time
mela:cat:health_centre	Health centres / स्वास्थ्य केंद्र	5 centres + current-shift staff
mela:cat:vet_camp	Veterinary camps / पशु चिकित्सा	11 camps + doctor + phone (no shifts)
mela:cat:parking	Parking / पार्किंग	18 locations (map link where lat/lon exist)

Row id	Title (en / hi)	Answer
mela:cat:ghat	Ghats / घाट	6 ghats (map link where lat/lon exist)
mela:rules	Do's & don'ts / क्या करें, क्या न करें	Hidden until <code>guidelines</code> has rows
mela:ask	Ask a question / सवाल पूछें	Next free text → LLM Q&A over <code>qa_context</code>

- **Card buttons:** thana, health, and vet cards end with [📍 Nearest] [🔄 All shifts] [🏠 Menu] (the 3-button maximum). `mela:roster:<category>` sends all 3 shifts; `mela:near:<category>` jumps straight to the location request.
- **Long messages:** any card over 4096 chars is split at place boundaries.
- **Free-text shortcuts:** the classifier maps "thana", "police", "doctor", "hospital", "parking", "ghat", "aaj ka program" and similar straight to these ids. A place name typed as text (e.g. "Kali Ghat") matches `places` by trigram and returns that one place's card.

6.1a "📍 Near me": nearest police / health centre / vet camp, with walking directions

Triggers

- Free text such as "police near me", "nearest hospital", "पास में थाना कहाँ है", or "dispensary kidhar hai". The classifier returns `mela.near{category}`.
- The 📍 Near me menu row.
- A card's [📍 Nearest] button.
- A citizen sharing a location with no pending question: the reply covers the nearest **thana + health centre + control room** together.

Flow

1. **Ask for location:** the bot sends `location_request` ("Tap *Send location* to find the nearest police station"). Session state becomes `mela.await_location{category}`, valid for 10 min. [Likely] The WhatsApp Cloud API supports this interactive type; if it is unavailable on the pinned Graph version, the bot falls back to the text instruction "📍 → Location → Send current location".
2. **Receive location:** the citizen's `location` message arrives, and the bot calls `nearest(category, lat, lon, 3)`.
3. **Reply:**
 - One text card with the 3 nearest sites. Each shows name, **≈ distance** (straight-line, labelled "approx."), the **in-charge on duty now** with 📞, and a 🗺️ **Walking directions** link: `https://www.google.com/maps/dir/?api=1&destination=<lat>,<lon>&travelmode=walking`.
 - The link deliberately has **no origin**, so Google Maps uses the phone's live GPS and keeps updating as the person walks. It also keeps the citizen's own coordinates out of any URL.
 - A native **location pin** for the nearest site; one tap opens it in the phone's Maps app.
 - Buttons: [🔄 Next 3] [📄 All <category>] [🏠 Menu].

Edge cases

Situation	Behaviour
Citizen is more than <code>mela_radius_km</code> from the Mela centre	"You seem to be outside the Mela area." The bot shows the full category list plus a directions link to the Mela control room
No site in the category has coordinates yet	"Location service for <category> starts soon." The bot shows the full list card (no dead end)
Citizen can't or won't share location (feature phone, GPS off, refuses)	"Or type a nearby landmark (e.g. Kali Ghat, Meena Bazar)." A trigram match on <code>places</code> uses that place's coordinates as the origin
Location shared while the session expected something else	Handled as "nearest of all key categories"

Situation	Behaviour
Two sites at about the same distance	Both are shown; the list is sorted by distance, then <code>sort</code>

Privacy:

- The citizen's latitude/longitude are **used in memory only and never stored**.
- `message_log` records `kind=location`, the category, and a distance bucket (<250 m, <500 m, <1 km, <2 km, more) for analytics.
- The privacy notice mentions location use.

6.1b Admin coordinate capture (on-site, in the set-up week)

Typing coordinates by hand into a sheet is error-prone (swapped lat/lon, missing decimals), and the temporary sites only exist once they are built. Staff therefore **pin each site by standing at it and sharing their WhatsApp location with the bot**.

1. An admin number sends `pin` (or taps `mela:adm:pin`), picks a category, then the site. The list shows sites **without coordinates first**, marked /, and is paginated.
2. The bot sends `location_request`; the admin shares the current location from the site's entrance.
3. The bot replies with the captured point as a map pin, plus  Save  Retry.
4. On Save:
 - the point is upserted into `place_coords`
 - it is written back to the sheet's `lat/lon` cells as best effort, which needs the service account to have Editor access on `places`
 - the event is logged with the admin's hash
5. `pin status` shows counts per category (e.g. "Thana 11/14 pinned"). The daily digest lists unpinned sites until they are all done.

Effort on the ground: about 50 sites. One or two staff on a bike can do it in about a day [Guessing], once the camps and thanas are physically up, typically in the week before the Mela opens.

7. Security & privacy

- **Network:** there are no inbound ports on the office machine. The Cloudflare Tunnel exposes only `/webhook/*` and `/healthz`. The n8n editor, Postgres, and Metabase are bound to 127.0.0.1 and reached over Tailscale.
- **On the fallback VPS:** `ufw` allows only 80/443 (none at all if the tunnel is reused). SSH is reachable only via Tailscale. `fail2ban` and `unattended-upgrades` are enabled.
- **Public vs personal data:** the Mela content is public-domain, which is why a private VPS is acceptable. Citizens' phone numbers and messages are personal data under the DPDP Act, so they stay hashed/encrypted, under 180-day retention and with no stored locations, **whichever host is used**.
- **Secrets:** kept in `.env` (chmod 600), never committed. Includes `N8N_ENCRYPTION_KEY`, `META_APP_SECRET`, `PHONE_HASH_SECRET`, and `PGCRYPTO_KEY`, all backed up offline.
- **Webhook authenticity:** the HMAC signature check (E2), plus the verify-token handshake.
- **DPDP Act 2023:**
 - **Notice:** a one-line privacy notice on first contact.
 - **Minimisation:** analytics use the hash only; the raw number is encrypted at rest.
 - **Retention:** 180 days, enforced by the nightly purge.
 - **LLM:** receives message text only.
 - **Location:** citizen coordinates are processed in memory and never stored or logged; only distance buckets are logged. Admin captures store the site's coordinates, not the person's.
- **Ethics:** the bot never asks for Aadhaar or other identity numbers.

8. Operations

- **Deploy:** `git pull && scripts/n8n_import.sh && psql -f sql/*.sql`. Credentials are created once per environment and documented.
- **Backups:**
 - nightly `pg_dump` of both DBs, 14-day local rotation plus an off-box copy
 - weekly copy of the env files and encryption keys to secure storage
 - one restore drill before go-live
- **Monitoring:**
 - an external uptime check on `/healthz` every 5 min
 - the Error Trigger → `admin_alert`
 - a sync-staleness check: no OK sync in 1h → alert
 - the daily report includes error count and p95 latency
- **docs/runbook.md:** covers restart, token rotation, "wrong number published" hotfix (edit sheet → sync or force-run), disabling a service, disabling the LLM, and rollback (re-import the previous git tag).

9. Testing strategy

Layer	Tool	What it covers
SQL	<code>tests/sql/*.sql</code> via <code>psql</code> + <code>pgTAP</code> -style asserts	<code>begin_turn</code> (dedupe, rate limit), <code>current_shift/on_duty</code> shift boundaries, <code>nearest()</code> (known points with hand-computed distances, inactive/unverified/no-coordinate sites excluded, sheet value overrides capture), <code>replace_*</code> atomicity (sync leaves <code>place_coords</code> intact)
Templates	<code>tests/check_templates.py</code>	WhatsApp field limits in hi and en, and no missing translations
Flow	<code>tests/run_flow_tests.py</code> + <code>tests/payloads/*.json</code> + <code>tests/services/<key>/*.yaml</code>	Real-shaped Meta payloads are sent to the staging webhook (mocks on). Asserts on captured outbound payloads, <code>message_log</code> , and <code>sessions</code>
Contract	same runner via <code>99-test-harness</code>	Output schema of every service for its standard inputs
LLM routing	<code>tests/llm/free_text.yaml</code> (50 hi/Hinglish/en lines, labelled)	Run against the real API; ≥90% correct service/subtype
Load	<code>tests/load_test.py</code>	6 msg/s sustained p95 < 1 s; 12 msg/s burst: 0 lost, 0 duplicates, self-recovery (docs/go-live.md §1)
UAT	<code>docs/uat-mX.md</code>	Real phones, every path, both languages

CI: GitHub Actions `ci.yaml` runs on every push:

1. Starts compose in `ENV=test` with the mocks.
2. Applies the SQL.
3. Imports the workflows.
4. Runs the SQL tests, template checks, and flow and contract tests.

The LLM-routing and load tests are run manually at their gates.

10. Milestones, tasks and estimates

Effort is in developer-days for 1 developer, plus a part-time data/ops owner on the client side.

Week-1 prerequisites (client side; these block go-live)

- ☒ WhatsApp number live and verified
- ☐ Meta: share the permanent System User token, Phone Number ID, **App Secret** (for signature checks); submit the `admin_alert` utility template; keep a separate **Meta test number for staging**
- ☐ Office machine meeting §2.2 (dedicated, Ubuntu-ready), UPS, 4G failover; it runs 24×7 from M1 for the burn-in (§2.4)
- ☐ **Hostinger (or similar) account** in the department's/owner's name, ready for the M2 drill and fallback (§2.4)
- ☐ **Named technical owner** authorised to trigger the VPS move (§2.4)
- ☐ A (sub)domain with its DNS on Cloudflare (free account); IT confirms outbound HTTPS to Cloudflare, Meta, Anthropic and Google is allowed
- ☐ Off-site backup target (a Google Drive folder or similar)
- ☐ Written approval for the hosted LLM (otherwise go live with `LLM_ENABLED=false`)
- ☐ Google service account; Mela sheet created and shared
- ☐ Named data owner for the Mela sheet
- ☒ Sample data received: `Sonpur_Mela_Data.xlsx` (7 tabs), plus the Directory Google Sheet (118 officers, kept for the later phase)
- ☐ **Public control room helpline number(s)** for the Mela (D8)
- ☒ Publishing duty-staff mobiles approved (D15): file a copy of the order
- ☐ Unicode spelling of "Nakhas" (the cell uses Kruti Dev encoding, D2); corrected numbers for D3/D4; confirm the D7 duplicates
- ☐ **Mela centre point and radius** (e.g. the Harihar Nath temple gate, 5 km) for the "outside the Mela area" check
- ☐ **Field staff (1–2 people, about 1 day) in the Mela set-up week** to pin every thana, health centre, vet camp, parking area and ghat with the admin tool (§6.1b); their numbers are added to `ADMIN_WA_NUMBERS`
- ☐ Service account given **Editor** access to the Mela sheet (for coordinate write-back)
- ☐ Optional: accommodation list, do's & don'ts, Hindi names
- ☐ The 2026 programme schedule, when announced (the sample is 2025)
- ☐ Later: **DM approval** to enable the Officer Directory (and afterwards Scheme Eligibility)

M0: Core platform: weeks 1–2, ~11 dev-days

Task	Days
Office machine setup: Ubuntu, Docker, systemd, Tailscale, cloudflared tunnel + access rules, UPS/boot settings	1
Compose rewrite: pinned versions, Postgres dbs, env, fixes E4–E6; staging + prod stacks	1
<code>00_core.sql</code> : tables, <code>begin_turn</code> / <code>end_turn</code> , views, seed services/templates	1.5
Router <code>00-router</code> : signature check, explode, dedupe, normalize, route, dispatch, end turn	2
<code>01-send</code> : renderer, limit validator, phone guard, retries, logging	1
<code>02-llm</code> : classifier + kill switch; <code>08-error</code> ; <code>03-admin-alert</code>	1
<code>svc_template</code> + <code>svc_echo</code> ; <code>99-test-harness</code>	0.5
Mocks, flow runner, template checker, CI	2
<code>n8n_export.sh</code> / <code>n8n_import.sh</code> , fixed IDs, runbook skeleton	1

Gate M0:

- CI green.
- Flow tests pass for: text, button, list, location, voice, status callback (no execution saved), batched webhook (both messages answered), duplicate ID (one reply), bad signature (rejected), stale button (correct), double-tap (consistent), rate limit.
- With only `svc_mela` enabled (via `svc_echo` in M0), "Hi" opens that service's menu directly. Enabling a 2nd service shows 2 buttons, and a 4th registry row flips the menu to a list.
- A fresh-instance import works with credentials as the only manual step.
- A real round-trip works on the Meta test number.

M1: Sonpur Mela: weeks 3–5, ~12.5 dev-days

Task	Days
<code>10_svc_mela.sql</code> : shifts, places, duty, control room, events, <code>current_shift/on_duty</code> , <code>replace_*</code> (the office machine starts the 24×7 staging burn-in, §2.4)	1
Template workbooks (dropdowns, text phone columns, README) + <code>scripts/convert_samples.py</code> (print layout → template: expand merged cells, split multi-phone cells, detect Kruti Dev text, flag bad phones) + conversion report + data-entry guide	2
<code>sync_mela</code> with validation (D2–D7, D11) and alerts/digest	1.5
<code>svc_mela</code> workflow: data-driven menu, today/schedule, control room on-duty, category cards with current shift, all-shifts roster, place-name lookup, ask (Q&A mode)	3
Fixtures (built from the real 2025 sample, re-dated) + SQL shift tests + 30-line LLM set	1.5
📍 Near me (§6.1a) : <code>nearest()</code> SQL, location request/receive, radius check, landmark fallback, directions link + map pin, card buttons; core send support for <code>location / location_request</code> messages	2
Admin coordinate capture (§6.1b) : <code>core.admins</code> , <code>mela:adm:*</code> flow, <code>place_coords</code> , sheet write-back, <code>pin status</code> , digest line	1
UAT on staging with the data owner (including a walk test at 3 real points with the admin tool and near-me)	0.5

Gate M1:

- All fixtures pass in hi and en.
- Shift tests pass: 05:59/06:00, 13:59/14:00, 21:59/22:00, and 02:00 (Shift-3 across midnight). Merged-cell desks show on every shift.
- The converter reproduces every source row: 5 health centres × 3 shifts, 14 thanas × 3, 11 vet camps, 18 parking, 6 ghats, 14 events, and 5 control desks.
- The converter report flags D2, D3, D4 and D7 items, and none of them reaches citizens.
- A broken sheet leaves the data unchanged and raises one alert.
- LLM routing ≥90%, with 0 invented numbers; the 30-line set includes 8 "near me" phrasings in Hindi, Hinglish and English.
- Near me: for 10 test origins, the returned top-1 matches a hand-checked nearest site. Directions links open walking navigation to the right point on Android and iPhone. Origins outside the radius, no-coordinate categories, and the landmark fallback behave as §6.1a says.
- Admin capture: a non-admin gets "not allowed" for `mela:adm:*`; a capture survives the next 10-min sync; a sheet-typed value overrides it.
- No citizen latitude/longitude appears in `message_log`, the n8n execution data, or the DB (grep check).
- The data owner signs off that the staging answers match the sheet.

M2: Go-live readiness: week 6, ~4 dev-days

Task	Days
k6 load test on the office machine (+ queue mode only if it fails)	1

Task	Days
Off-site backups + <code>scripts/provision_vps.sh</code> + restore drill onto a throwaway Hostinger-type VPS (\$2.4)	1.5
UptimeRobot review, power-cut / ISP-failover test, runbook complete, hosting go/no-go against H1-H6 recorded in <code>docs/hosting-decision.md</code>	0.5
Prod stack + production number cut-over, UAT with about 20 staff phones, Hindi copy review	1

Gate M2:

- Load test meets the targets. They were revised after measurement (`docs/go-live.md` §1): sustain **6 msg/s with p95 < 1 s** on the target machine, and survive **12 msg/s for 30 s** with 0 lost, 0 duplicates and self-recovery. The original "30 msg/s" was never derived from expected traffic; the expected Mela peak is about 2 msg/s, with bursts of about 6.
- The restore drill onto a fresh VPS completes in under 1 hour.
- **Hosting go/no-go**: all of H1-H6 pass → go live on the office machine. Any fail → go live on the VPS (allow 1 extra day).
- Pulling the power cable and the primary internet cable: the bot recovers without anyone touching it (H3/H4).
- UAT sign-off.
- **Go-live around 10 Nov**, with the prod webhook switched to the production number.

M3: Analytics & feedback: weeks 7-8 (during the Mela), ~5 dev-days

Task	Days
Metabase container + read-only DB user on views	1
Dashboards: daily users/messages, per service/subtype, top free-text themes, 🗨 rate, unanswered list, peak hours, LLM tokens/cost	2
85-daily-report at 08:00 IST via <code>admin_alert</code> + email; 09-purge job	1.5
Weekly <code>unanswered</code> review → keyword/menu/data fixes	0.5 (+ ongoing)

Gate M3:

- A sampled day reconciles with `message_log`.
- The purge is verified on a copy.
- No raw numbers are visible anywhere in Metabase.

M3 as built (code complete; each gate item has an automated check):

- **Logging gaps closed first:**
 - Q&A tokens are now counted.
 - Near me logs a distance bucket and origin (`gps:<500m, landmark:...`), never the location.
 - Q&A "don't know" answers go into `core.unanswered`.
- **Privacy rules in the analytics schema:**
 - Citizens appear only as a 10-character ref, and typed digit runs and e-mails are masked.
 - Admin phones and pre-go-live (`analytics_since`) rows are excluded.
 - The Metabase login `metabase_ro` can read only this schema.
- **Gate evidence:**
 - Reconciliation: `analytics.reconcile(day)` is covered by the SQL tests and flow tests.
 - Purge on a copy: `tests/restore_drill.sh` runs `core.run_purge()` on the restored copy.
 - No numbers in Metabase: `scripts/metabase_setup.py --verify` runs every card, scans the values and probes `core`. It also runs in CI against a real Metabase 0.63.
- **Daily report:**
 - WhatsApp `admin_alert` gets one line, because template parameters can't contain newlines.
 - SMTP e-mail gets the full text. It is optional, because the template may not be approved in time.

- Services add their own line through `svc_<key>.digest()`; Mela reports map-pin coverage and sheet-sync freshness.
- **Daily totals are archived before each purge** (`analytics.daily_archive`), so next year's Mela can be compared with this one after the 180-day deletion.
- **"Top free-text themes"** is a normalised-text grouping of unanswered questions, not LLM clustering. [Likely] It is enough at Mela volumes and costs nothing. Revisit if the list gets long.

Phase 1 total: about 32.5 developer-days (M0 11 + M1 12.5 + M2 4 + M3 5). The go-live path (M0–M2) is about **27.5 days (≈5.5 weeks)**, which leaves **about 1 week of buffer** before ~10 Nov.

Slip rule (less likely to be needed now): if M1 runs late, 📍 Near me and admin capture are the first items to move. They can ship during the first Mela week without affecting anything else, because the feature stays switched off until coordinates exist anyway. Coordinates are captured in the set-up week, so the feature realistically goes live **when the camps are up (around 17–23 Nov)** whichever way the schedule falls.

11. Risk register

Risk	Likelihood	Impact	Mitigation
admin_alert template not approved	Low	No WhatsApp staff alerts	Email/SMS alerts as backup
LLM approval refused	Medium	Weaker free-text handling	<code>LLM_ENABLED=false</code> still covers every service via menus
Office machine / power / internet outage during the Mela	Medium-High	Bot silent	\$2.2 hardening; 4-week burn-in measured against H1–H6 → go/no-go at M2; Hostinger-type VPS standby with a rehearsed 1-hour move; mid-Mela triggers for a same-day move (\$2.4)
Office IT blocks Cloudflare/Anthropic egress	Medium	Can't go live on-prem	Confirm in week 1 (H6); go live on the VPS instead
Mela data not verified in time	High	Wrong numbers published	Named owner, <code>verified</code> column, prod rejects unverified rows, sign-off in the M1 gate
Duty roster changes mid-Mela not updated in the sheet	Medium	Wrong on-duty contact	The daily digest shows the age of <code>verified</code> ; the data owner updates it by 10 am; <code>publish_duty_phones</code> switch
Traffic spike during the Mela	Low–Medium	Slow replies	Load test on the actual office machine; queue-mode profile ready
Site coordinates not captured, or captured wrongly	Medium	Near me unavailable, or it points to the wrong spot	Admin tool with map-pin confirmation; <code>pin_status</code> in the daily digest; the feature hides categories without coordinates; a sheet value can correct any point
Citizens' GPS inaccurate in dense crowds	Medium	"Nearest" off by 100–200 m	Show the top 3 with "approx." distances; the walking link recalculates live in Maps
Citizens send voice notes	High	Unanswered	Friendly redirect to the menu; log volume; voice-to-text as a future service
n8n upgrade breaks nodes	Low	Outage	Pinned versions; upgrade only through staging + CI

12. Running cost estimate (Mela month) [Guessing: volumes are assumptions]

- **Hosting:**

- **Office machine:** existing, or about ₹60–80k one-time if bought, plus a UPS.
- **Connectivity:** a 4G backup plan at about ₹300–500/month; Cloudflare Tunnel is free; the domain is about ₹800/year.
- **Fallback VPS** (Hostinger-type): about ₹700–1,500/month **only if activated**, plus about ₹100–200 for the M2 drill.
- **WhatsApp:** [Likely] free for replies to citizen-initiated chats (the service window). Utility templates for admin alerts cost a few paise to about ₹0.15 each.
- **Claude Haiku 4.5** (\$1 input / \$5 output per MTok):
 - Assume 50k free-text calls × ~1.5k input + 200 output tokens.
 - That gives ≈ \$75 input + \$50 output, about **\$125 (≈₹10.5k)** for the whole Mela.
 - Button taps cost nothing because they never call the LLM.

13. Immediate next steps

1. **District / client side:** close the week-1 prerequisites in §10. The ones that most affect the schedule are:
 - the office machine (and its 24×7 burn-in from M1) and the Cloudflare domain
 - the Meta App Secret and a test number
 - the public control room number
 2. **Development:** start M0 (core platform) on the staging stack as soon as the office machine is available. Until then, development runs on a laptop with the same Docker setup.
 3. **Data owners:** receive the template workbooks and the data-entry guide in M1, and start moving the 2026 duty orders into them as soon as the orders are issued.
-

14. Later phases (out of Phase 1 scope)

How any new service plugs in (no router change)

1. Copy `svc_template` and give it a new fixed workflow ID.
2. Add a new `svc_<key>` DB schema.
3. Write fixtures and pass the contract test.
4. Insert one `core.services` row and its templates.
5. Enable it on staging, run UAT, then enable it on prod.

The main menu grows automatically. In Phase 1 (one service), "Hi" opens the Mela menu directly. With 2–3 services it shows reply buttons, and with more than 3 a list.

Officer Directory (next phase, after DM approval): design ready

Effort: about 7 dev-days.

- 1.5 d: schema, sheet, and sync
- 3 d: the service
- 1 d: fixtures
- 1 d: UAT and go-live
- 0.5 d: converting the sample

Scope agreed so far: Saran district, sub-division, and block level, including **all block-level posts** (not only BDO/CO), plus police stations.

Needs before start: DM approval, block-level officer lists for all 20 blocks, and a named data owner.

Gate: 30 real-style queries resolve to the right officer in ≥90% of cases ("BDO Dighwara", "Sonpur CO", "ration card complaint", "DM number", "Derni thana"), **with no core feature change**.

Data model

svc_directory schema (20_svc_directory.sql), shaped by the real sample

- **officers** (id, level, unit_en, unit_hi NULL, designation_en, designation_hi NULL, person_name, phone NULL, email NULL, keywords text[], sort, last_verified).
 - level is one of: district, subdivision, block, circle, police.
 - unit_en holds the block, circle, or thana name ("Amnour"), the subdivision ("Sonpur"), or "Saran" for district level.
- **blocks** (name_en, name_hi, aliases text[]): the 20 Saran blocks. Aliases include Sadar = Chapra, Rivilganj = Revilganj, and Isuapur = Ishuapur.
 - Used for the paginated list and fuzzy matching.
- **Indexes:** trigram indexes on unit_en, designation_en, person_name, and blocks.aliases.
- **Function:** replace_officers(jsonb).

Findings from the directory sample

#	Finding	Evidence	Consequence / action
D12	Police thanas aren't linked to blocks (39 district thanas vs 20 blocks). Sub-division rows all read "Subdivisional Officer" and only the email reveals Chapra/Marhaura/Sonpur	Directory sheet	A "block → department" flow doesn't fit. The directory is navigated by office level first (§6.2). Converter sets unit_en for sub-divisions from the email
D16	Police display names carry disambiguation suffixes ("Randhir Kumar-2", "Mukesh Kumar-01")	Directory Thana section	The display strips a trailing -\d+; the stored value is kept as-is

Sheet format

Spreadsheet: "Saran Officer Directory – Bot Data"

Tab	Columns	One row =
README	rules	n/a
officers	id, level ▼ (district, subdivision, block, circle, police, other), unit_en, key (TRUE = show on the block summary card), unit_hi, designation_en, designation_hi, short_title_en (≤24 chars), person_name, phone, phone_2, email, office_address, keywords (comma-separated citizen words, e.g. ration, राशन, PDS), sort, active, last_verified, verified, updated_by	one post. Transfers are made by editing the name/phone on the same row , never by adding a row
units	unit_en, unit_hi, level ▼, aliases (e.g. Sadar, Chapra, छपरा)	one block, circle, subdivision or thana name, used for search and menus

Why keywords matters: citizens ask by need ("pension", "zameen", "ration card"), not by designation. A data owner who fills keywords improves search more than any code change does. The LLM only falls back to it when keywords miss.

Service design

svc_directory (id_prefix dir). Navigation by office level, matching the data (D12)

Level menu (list):

- dir:lvl:district: District officers (33)
- dir:lvl:subdivision: Sub-division officers (SDO/DCLR, 6)
- dir:lvl:block: Block officers (BDO, CO, and all other block-level posts)

- `dir:lvl:police`: Police stations (39)
- `dir:search:`  Type a name/post

Block (`dir:blk:p1, p2`; 20 blocks shown as 9 per page + "More ►"):

- Tapping a block (`dir:blk:<name>`) returns a **key-officers card** (BDO + CO + the posts marked `key=TRUE`, e.g. MOIC, BEO, CDPO), since citizens rarely know which one they need. It ends with `[ All officers of <block>]`.
- `dir:blkall:<block>;p1` shows every block-level post as a paginated list: BAO/BHO, BEO, MOIC, CDPO, BCO, MO (Supply), JE/AE, Panchayat Raj Officer, Labour Enforcement Officer, BWO, BPM (JEEViKA), RO, and others. Tap → the officer card.
- **Search is the main path for these posts.** "Dighwara ka hospital doctor" → MOIC Dighwara. "anganwadi Sonpur" → CDPO Sonpur. It works through `keywords` plus the LLM need → designation mapping.

District (33 posts):

- A paginated list of designations. Titles are shortened by a `short_en` rule (e.g. "District Supply Officer" → "Supply Officer").
- Tap → officer card.

Sub-division: Sadar / Marhaura / Sonpur → a card with the SDO + DCLR.

Police (39 stations): a paginated list (5 pages), or type the thana name (trigram).

Search / free text, e.g. "BDO Dighwara", "ration card officer", "DM ka number":

1. Trigram match on unit, designation, and name (+ aliases).
2. If there is no confident match, the LLM maps the need to a designation from the 33+ known posts via the `keywords` column. For example, ration card → District Supply Officer; land → CO/DCLR; pension → ADSS Social Security.
3. The bot shows the top 1 match, or up to 3 as buttons.

Officer card: designation, name (suffix stripped),  phone,  email (omitted if blank), unit, "Verified: <date>".

Scheme Eligibility (after the Directory): notes kept for later

- **Data source:** structured MySQL columns, exposed through a read-only view and synced nightly into its own `svc_schemes` schema.
- **Verdict:** decided by a deterministic SQL function, never by the LLM.
- **Flow:** about six button/list questions, each with a skip option.
- **Output:** each result gives the reason and an "indicative only" disclaimer.
- **Effort:** about 10 dev-days.
- **Needs:** a MySQL schema sample and department-reviewed test profiles.